

Data Cleaning and Quality Assessment

Homework 3 – Repurposing data sources

Background

The COVID-19 pandemic resulted in significant health, economic, and social impacts worldwide. Governments launched large-scale vaccination campaigns to reduce infections, hospitalizations, and deaths. Analyzing vaccination rollout information provides insight into public health policy, communication strategies, and data reporting practices. In this assignment, you will extract and compare information about the COVID-19 vaccination rollout in [United States](#) and [Australia](#). You will perform the extraction twice: once from the HTML pages and once from the corresponding PDF documents.

1 Extracting information from HTML documents (40 pts)

Use web-scraping techniques covered in this course to extract structured information from the two Wikipedia articles.

- a) Download and parse the HTML documents.
- b) Extract the table(s) with the vaccination rollout.
- c) Store the extracted content in a structured format (e.g., CSV, JSON, or SQLite database).
- d) Compare the extracted information between the United States and Australia.

2 Extracting information from PDF documents (40 pts)

Compare PDF-based information extraction with HTML-based extraction.

- a) Obtain the PDF version of each article (print the page to PDF, or download the PDF version available through Wikipedia).
- b) Extract the table(s) with the vaccination rollout.
- c) Convert the extracted information into a structured format.
- d) Compare the PDF extraction results with those obtained from HTML.

3 Analysis questions (20 pts)

- a) Which format was easier to parse: HTML or PDF?
- b) Which format preserved tables more accurately?
- c) What information was lost or distorted during PDF extraction?
- d) What preprocessing steps improved extraction quality?
- e) Which format would you recommend for large-scale data collection and why?

Extra credit (10 pts): Visualization and insight generation

Using the extracted data, create **two visualizations** that compare the vaccination rollout in the United States and Australia.

Examples:

- Vaccination timeline comparison
- Vaccines approved over time
- Milestone comparison (first dose, fully vaccinated, booster rollout)
- Word cloud of major public-health topics discussed in the articles

Provide a short discussion explaining:

- Why you selected the visualizations?
- Key insights revealed by the analysis.
- Which country's rollout information was easier to interpret from the extracted data?

Submit

Upload to Canvas your solution, in a **.zip** file that includes the following:

- Everything needed to make your work reproducible (e.g., **.Rmd**, **.ipynb**, etc.).
- The **.pdf** with your answers and the visualizations. Make sure you indicate which answer corresponds to which question (i.e., create section headers in the Jupyter Notebook or **.Rmd** file).